

SONAR-X

Physics-Aware Explainable Marine Debris Intelligence System

SURVEY INTELLIGENCE REPORT

■ DATA SOURCE: DEMO — Synthetic data — not field performance. All results are demonstration outputs.

Survey Information

Survey ID	SX-SRV-DEMO-001
Survey Name	Arabian Sea — Demo Survey
Survey Date	2024-03-15
Area	Arabian Sea — Western Coast
Operator	Team Kurukshetra
Data Source	DEMO
Status	COMPLETED
Generated	2026-09-24 14:09 UTC
Software	SONAR-X 0.1.0-prototype

Sonar Metadata

Altitude above seabed	2.5 m
Range scale	0.2 m/px
Channel layout	BOTH
Already gain corrected	No
Already slant-range corrected	No

Detection Summary

Total detections	5
Marine Debris	1
Natural Formation	1
Sonar Artifact	1
Unknown Anomaly	2
Hazard — High Caution	1
Hazard — Caution	1

Detection List

Detection UID	Class	Conf.*	Evidence*	Risk	Hazard	Lat	Lon	Status
SX-DET-SX-SRV-DEMO-001-0005	UNKNOWN ANOMALY	64	72	MEDIUM	HIGH_CAUTION	19.4530	72.8445	PENDING
SX-DET-SX-SRV-DEMO-001-0004	UNKNOWN ANOMALY	41	45	MEDIUM	CAUTION	19.4510	72.8418	PENDING
SX-DET-SX-SRV-DEMO-001-0003	SONAR ARTIFACT	85	82	LOW	NONE	N/A	N/A	PENDING
SX-DET-SX-SRV-DEMO-001-0002	NATURAL FORMATION	72	68	LOW	NONE	19.4498	72.8435	PENDING
SX-DET-SX-SRV-DEMO-001-0001	MARINE DEBRIS	78	81	LOW	NONE	19.4521	72.8403	CONFIRMED

** Confidence and Evidence Score are prototype scores, NOT validated probabilities.*

■ HAZARD ASSESSMENT

The following detections triggered precautionary hazard flags. This system does NOT identify specific hazardous objects. "Potential Hazard — Do Not Disturb" means sonar evidence warrants human review. Absence of a hazard flag does NOT mean an object is safe.

● **SX-DET-SX-SRV-DEMO-001-0005 — HIGH_CAUTION — UNKNOWN_ANOMALY**

Indicators: Large spatial extent, Regular / structured geometry, Strong acoustic backscatter, Prominent acoustic shadow, Unknown class with strong acoustic evidence

● **SX-DET-SX-SRV-DEMO-001-0004 — CAUTION — UNKNOWN_ANOMALY**

Indicators: Unknown class with ambiguous acoustic evidence

Limitations and Disclaimers

- DATA SOURCE: DEMO — Synthetic data — NOT field performance.
- This is a prototype system. All detections require human verification before any action.
- Confidence and evidence scores are prototype heuristic values, NOT validated probabilities.
- Side-scan sonar does not directly provide water temperature or depth.
- Conventional side-scan sonar is not a direct bathymetric sensor.
- Slant-range correction geometry is approximate where metadata was unavailable.
- Geolocation is approximate. GPS coordinates require field calibration.
- A missing detection does NOT prove debris was removed or moved.
- Absence of a hazard flag does NOT mean an object is safe.

AI-generated observations require verification by qualified human experts.